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C

$$\begin{aligned}\Delta x &\geq \frac{h}{\Delta p} \\&= \frac{h}{0.01p} \\&= \frac{h}{0.01\sqrt{2m_p E}} \\&= \frac{6.63}{0.01\sqrt{2(1.67 \times 10^{-27})(1.00 \times 10^6 \times 1.60 \times 10^{-19})}} \\&= 2.87 \times 10^{-12} \text{ m}\end{aligned}$$

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A